



AGH UNIVERSITY OF CRACOW

Project documentation - Design Lab 2025

Authors:

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1. Project description

This repository contains laboratory and final project work created during **Design Lab** classes. The project focuses on building an embedded IoT system using the **ATMega328PB** microcontroller and multiple communication interfaces.

2. Features:

The projects support various connectivity protocols such as:

- UART
- I2C
- SPI
- 1-WIRE
- Outputs (7seg display)

Software functions

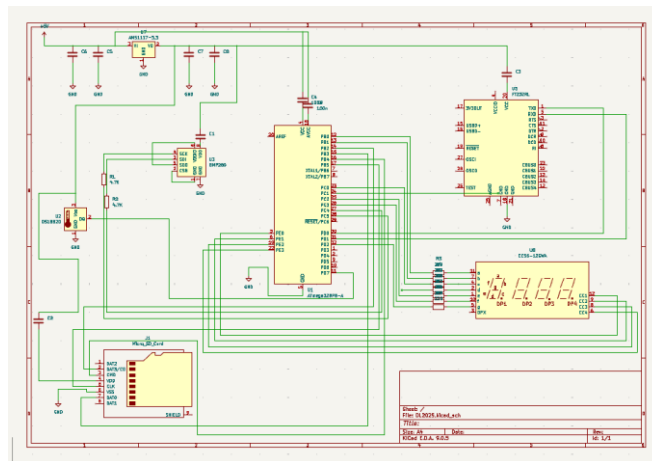
- UART connectivity and displaying counter of 7-seg display as well as data obtained from the sensors.

3. Hardware:

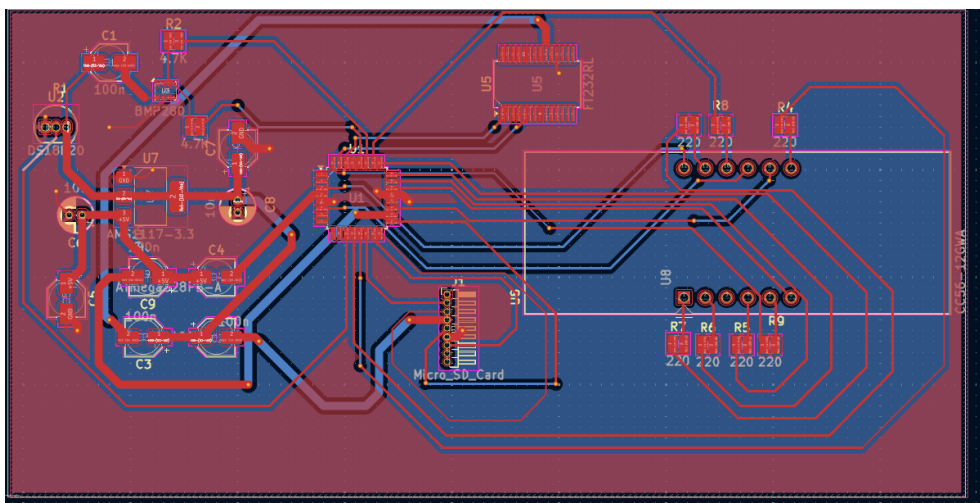
- ATMega328PB - microcontroller
- BMP280 - Pressure sensor
- DS18B20 – temperature probe
- PL2303 - UART connectivity
- 3.3V voltage stabilizers
- Capacitors (100n ceramic and 10uF electrolytic)
- Resistors (4,7K to pullup, 220 for 7seg)

4. Schematics:

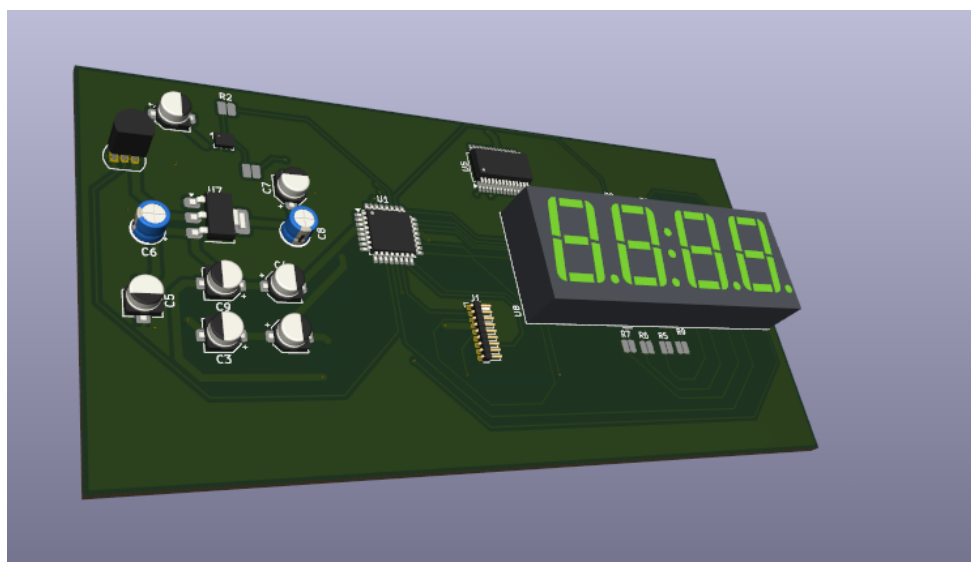
Schematic:



PCB layout:



3D model of layout:



5. Project structure:

- /datasheets/- Components datasheets
- schematics/ - Schematics and PCB designs

- I2C-BMP280/ - I2C examples with BMP280
- Final Project – complete DL code with support for all protocols
- README.md

6. Requirements

KiCad is required to open the schematics.

To build and run firmware you need:

- Software to program ATmega like AVR Microchip studio
 - AVR_GCC toolchain
 - “avr/io.h” , “util/delay.h” , “stdint.h” , “avr/interrupt.h” libraries
- If you want to compile project via terminal use these commands:
avr-gcc -mmcu=atmega328pb -Os -o main.elf source_file.c
avr-objcopy -O ihex main.elf main.hex

To flash use:

- *avrdude -c usbasp -p m328pb -U flash:w:main.hex*

Change settings/filenames if needed.

7. Connectivity with board

Data can be obtained via USB to Uart converter or via serial port. The circuit sends data which can be saved and later analyzed.

8. Possible Future improvements

Include saving data on SD card via SD card reader. Developing connectivity with the board via Bluetooth or Wi-Fi module. Creating a pc/mobile/web app displaying the measurements and options to save the data.

9. Links/sources

GitHub - <https://github.com/granovskyyy/DL2026>

10. Notes

This project is for educational purposes only.